Didier Izere

Cybersecurity Researcher | Post-Quantum Cryptography | Cybersecurity
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SUMMARY

Cybersecurity researcher pursuing MSIT at Carnegie Mellon University Africa, focused on post-quantum cryptography (PQC) readiness. Certified Ethical Hacker with experience in vulnerability assessment, network security, and cybersecurity education. Researching Rwanda's policy preparedness for the quantum transition, including Harvest Now, Decrypt Later (HNDL) risk and PQC migration. I am interested in cryptographic policy and cybersecurity capacity building in Africa.

EDUCATIONAL BACKGROUND

Master of Science in Information Technology **August 2026 – May 2028**
Carnegie Mellon University, Africa
Concentration: Cybersecurity

Bachelor of Science in Information Management **January 2020 – November 2024**
Adventist University of Central Africa, Kigali, Rwanda
Concentration: Information Management

PROFESSIONAL EXPERIENCE

IT Assessor /Teacher (BTEC Curriculum) **January 2025 – August 2026**
King David Academy, Kigali, Rwanda

- Assessing BTEC IT coursework and practical projects against Pearson standards
- Mentoring students in networking, cybersecurity, and project management.
- Ensuring assessment quality and compliance with academic standards
- Collaborating with staff to enhance curriculum relevance

Head of ICT Department & IT Teacher (Cambridge Curriculum) **August 2024 – January 2025**
Mother Mary International School Complex, Kigali, Rwanda

- Taught Cambridge IT curriculum to candidate classes (Grades 11 and 12), preparing students for external examinations
- Supervised ICT staff and delivered practical training in systems and security
- Introduced digital tools that improved teaching and administration efficiency
- Organized ICT workshops for staff and students

Cybersecurity Intern **June 2023 – November 2023**
ITEC LTD, Kigali, Rwanda

- Conducted vulnerability assessments and cybersecurity risk analyses for client organizations
- Implemented security controls and recommended mitigation strategies
- Assisted in developing cybersecurity awareness and client training programs

PROJECTS

Post-Quantum Cryptography Readiness Assessment
July 2026 – Present

- Analyzing Rwanda's financial sector cybersecurity standards for PQC readiness
- Comparing national policy frameworks against international PQC migration guidance
- Identifying regulatory gaps in the financial sector's preparedness for quantum threats
- Developing an empirical framework to assess policy-level PQC readiness

Tekmax Cyber Range
May 2026 – Present

- Building browser-based cybersecurity training platform for Rwandan secondary schools
- Developing localized scenarios including mobile money fraud, SIM swap, and government domain spoofing
- Creating hands-on labs for packet analysis, web security, and incident response
- Designing student progress tracking and verifiable skill certifications

CERTIFICATIONS

- **Certified Ethical Hacker (CEH)** – EC-Council
- **Cybersecurity Engineering Virtual Internship** – SAP
- **Internet Governance** – Internet Society
- **NDG Linux Essentials** – Cisco Networking Academy
- **Network Security** – Cisco Networking Academy
- **Introduction to Cybersecurity** – Cisco Networking Academy

SPECIALIZED TRAINING

- **CryptoHack – Introduction to Cryptography and Modular Arithmetic**
- **Cybersecurity Essentials** – CyberHub, Rwanda National Cybersecurity Authority

SKILLS

- **Programming**: Python, Php
- **Operating Systems**: Linux
- **Web Technologies**: HTML, CSS, JavaScript
- **Cybersecurity**: Vulnerability Assessment, Network Security, Web Application Security, Cryptography, TLS Security
- **Security Tools**: Wireshark, Burp Suite, Metasploit, Nmap, Nessus

INTERESTS & ACHIEVEMENTS

- **picoCTF 2023** – 1st Runner-up, Carnegie Mellon University Africa
- **GCI World 2026** – Selected participant, Matsuo-Iwasawa Laboratory, University of Tokyo
- **Quantum Summer Academy 2026** – Selected participant, CMU Center for Quantum Computing and Information Technologies

PROFESSIONAL AFFILIATIONS

- **IEEE CMU-Africa Student Branch** – Active member
- **IEEE QUANTUM** – Active member
- **Post-Quantum Cryptography Coalition (PQCC)** – Active member contributing to global PQC adoption discussions
- **African Quantum Consortium (AQC)** – Active member engaging with Africa's quantum research community
- **Internet Society Rwanda Chapter** – Active member